PINNs for Options Pricing

FYS5429 – Advanced Machine Learning and Data Analysis for the Physical Sciences

STATUS REPORT

Egil Furnes
egilsf@uio.no
github.com/egil10/fys5419

University of Oslo
Department of Physics
March 18, 2026

STATUS REPORT AND PROGRESS

This section summarizes the current state of the project and outlines the planned work ahead. All source code is maintained at (Furnes 2026a) and the LaTeX source of this document at (Furnes 2026b).

Work Completed

A substantial portion of the early project work has gone into reading, orientation, and conceptual groundwork. Coming from a background in finance, a central early question was one of feasibility: whether a physics-informed approach could handle the specific structure of financial PDEs in a way that adds genuine value over existing analytical or numerical methods.

This led to a careful study of the model hierarchy underlying the project. The Black–Scholes model, despite its well-known limitations, remains the natural entry point: its PDE has an exact analytical solution, which makes it an ideal testbed for verifying that a PINN solver learns the right structure.

Moving from constant to stochastic volatility via the Heston model introduces a two-dimensional, time-dependent PDE with a coupled variance process and no simple closed form—a setting where mesh-free, data-driven methods become genuinely attractive. The natural further extension is the inverse problem: rather than pricing given a fixed set of parameters, one asks what parameter values are consistent with observed market prices. This is the calibration problem, and it is here that the PINN framework may offer the most practical value.

In parallel with this conceptual work, exploratory code has been written to generate option price surfaces under both models, experiment with collocation point sampling, and build intuition for the relevant parameter regimes. The introduction and methods sections of this document reflect the theoretical synthesis carried out during this phase.

Planned Work

The remaining work follows naturally from the model hierarchy established above.

Phase 1 — Black–Scholes Forward Problem

Implement and validate a PINN solver for the one-dimensional Black–Scholes PDE. The network is trained on collocation points in the (S, t) domain, with the terminal payoff and boundary conditions enforced through the composite loss in eq. (6). Prices and first-order Greeks (Δ, Θ) are compared against the analytical solution as a primary quality check.

Phase 2 — Activation Function Study

Systematically compare smooth activation functions (`tanh`, Swish, GELU, Softplus, SIREN) on the Black–Scholes problem, evaluating convergence speed and accuracy of second-order Greeks (Γ, Vomma). This phase is primarily experimental and is expected to yield concrete guidance on architecture choices for the more demanding Heston problem.

Phase 3 — Heston Forward Problem

Extend the PINN framework to the two-dimensional Heston PDE over (S, v, t). The additional complexity of the mixed derivative term $\partial^2 V/\partial S\,\partial v$ and the Feller boundary condition on the variance process is addressed here. Results are compared to a reference finite-difference or Monte Carlo solution.

Phase 4 — Inverse Problem and Calibration

Formulate the calibration problem as an inverse PINN: infer the Heston parameters $(\kappa, \theta, \xi, \rho, v_0)$ from observed option prices by augmenting the PINN loss with a data-fit term. Experiments are run on both synthetic and, if feasible, real market data.

Phase 5 — Write-up and Submission

Complete the results section with figures and tables. Write the conclusion, connecting findings back to the research questions. Final proof-read, bibliography check, and submission.

Key Risks

Training stability is the main practical concern: PINN losses for financial PDEs can be poorly conditioned due to the wide range of option values across the domain. Loss weighting and adaptive sampling strategies may be required. The calibration phase is the most experimental part of the project and may be simplified if the forward Heston solver proves insufficiently accurate.

Note: Below is the current template and outline of the report, with more to come.

Abstract

This project investigates the use of Physics-Informed Neural Networks (PINNs) for solving and calibrating option pricing models in quantitative finance. We begin with the Black–Scholes partial differential equation (PDE), demonstrating how its equivalence to the heat equation motivates a PINN-based solver that embeds the governing differential operator directly into the training loss. The neural network is trained to satisfy both the PDE residual and the terminal payoff condition, and its prices and Greeks are compared against the known analytical solution. We then extend the framework to the Heston stochastic volatility model, which introduces a higher-dimensional PDE with a mixed derivative term and no simple closed-form solution. The PINN approach is shown to scale naturally to this setting. We further explore the use of PINNs for the inverse calibration problem: inferring Heston model parameters from observed option prices. The results illustrate the promise of physics-informed machine learning as a flexible, mesh-free solver for financial PDEs, while also exposing practical challenges related to training stability and loss balancing in multi-dimensional settings.

1 Introduction

The classical starting point for many partial differential equation (PDE) methods in mathematical finance is the heat equation, which describes the diffusion of temperature in a homogeneous medium. In one spatial dimension, the heat equation is given by

$$\frac{\partial u}{\partial t}(x,t) = \alpha \frac{\partial^2 u}{\partial x^2}(x,t),$$

where $u(x,t)$ represents temperature, t denotes time, x is the spatial coordinate, and $\alpha > 0$ is the thermal diffusivity constant. The equation states that the rate of change in temperature is proportional to the spatial curvature of the temperature field, capturing the fundamental mechanism of diffusion.

The importance of the heat equation in option pricing arises from the fact that the Black–Scholes PDE can be transformed into a diffusion equation of this form through a suitable change of variables.

Under the standard geometric Brownian motion model for an asset price S_t,

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

the value $V(S,t)$ of a European option satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

By introducing the log-price transformation

$$x = \ln\left(\frac{S}{K}\right), \quad \tau = T - t,$$

together with an exponential rescaling of the dependent variable, the Black–Scholes equation can be reduced to the standard diffusion equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

This connection provides both analytical insight and a natural bridge to modern numerical approaches. In this thesis, Physics-Informed Neural Networks (PINNs) are used to solve the Black–Scholes PDE by embedding the governing differential equation directly into the loss function, allowing neural networks to approximate option prices while respecting the underlying diffusion structure inherited from the heat equation.

While the Black–Scholes model assumes constant volatility, empirical financial markets exhibit stochastic volatility dynamics. A widely used extension is the Heston stochastic volatility model, in which both the asset price and its variance evolve as stochastic processes:

$$dS_t = rS_t dt + \sqrt{v_t}\, S_t dW_t^S,$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}\, dW_t^v,$$

with correlation

$$dW_t^S dW_t^v = \rho dt.$$

In this setting, the option price becomes a function $V(S,v,t)$ and satisfies the two-dimensional Heston PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\xi vS\frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\xi^2 v\frac{\partial^2 V}{\partial v^2}$$
$$+ \quad rS\frac{\partial V}{\partial S} + \kappa(\theta - v)\frac{\partial V}{\partial v} - rV = 0.$$

Compared to the Black–Scholes equation, the Heston PDE introduces an additional spatial dimension and a mixed derivative term, making classical numerical solution methods more computationally demanding. This increased complexity makes the model particularly well-suited for physics-informed machine learning approaches. In this thesis, PINNs are therefore applied not only to the Black–Scholes equation but also to the Heston PDE, demonstrating how neural-network-based solvers can handle higher-dimensional option pricing problems governed by diffusion-type equations.

1.1 The Black–Scholes Model

The Black–Scholes model (Abdelmessih 2026) prices a European call or put option analytically under the assumption of constant volatility. Given the log-moneyness

variables d_1 and d_2,

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

the call and put prices are

$$C = SN(d_1) - Ke^{-rT}N(d_2),$$

$$P = Ke^{-rT}N(-d_2) - SN(-d_1),$$

where S is the current asset price, K the strike, r the risk-free rate, σ the volatility, T the time to maturity, and $N(\cdot)$ the standard normal CDF.

The Greeks

The sensitivity of the option price to its inputs — the Greeks — are partial derivatives of V with respect to market and model parameters. The first- and second-order Greeks are summarized in table 1, together with selected third-order Greeks. The Black–Scholes PDE can be written compactly in terms of the first- and second-order Greeks as

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0,$$

which makes explicit that the time decay Θ is balanced by gamma and delta exposure.

Table 1. Selected Black–Scholes Greeks.

Greek	Expression	Order
Δ	$\partial V/\partial S$	1st
Θ	$\partial V/\partial t$	1st
ρ	$\partial V/\partial r$	1st
Φ	$\partial V/\partial q$	1st
Vega	$\partial V/\partial \sigma$	1st
Γ	$\partial^2 V/\partial S^2$	2nd
Vomma	$\partial^2 V/\partial \sigma^2$	2nd
Vanna	$\partial^2 V/\partial S\,\partial \sigma$	2nd
Charm	$\partial^2 V/\partial S\,\partial t$	2nd
Speed	$\partial^3 V/\partial S^3$	3rd
Zomma	$\partial^3 V/\partial S^2\,\partial \sigma$	3rd
Color	$\partial^3 V/\partial S^2\,\partial t$	3rd

2 Methods

This section develops the methodological framework for the project. We first review the connection between maximum likelihood estimation and mean squared error as the natural training objective for neural networks. We then describe the Physics-Informed Neural Network (PINN) architecture and its application to the Black–Scholes and Heston PDEs.

Modern quantitative finance sits at the intersection of stochastic modeling, numerical methods, and machine learning. Classical stochastic volatility modeling builds on multiscale diffusion techniques, where volatility evolves on multiple time scales (Fouque et al. 2011). Analytical approaches to option pricing under stochastic dynamics are further developed in tractable model frameworks (Gulisashvili 2012), while probabilistic interpretations of derivative pricing link option values to risk-neutral expectations (Prefetta n.d.). Foundational deep learning theory and architectures are presented in Goodfellow et al. (2016), while practical implementations using modern software frameworks are discussed in Raschka et al. (2022). Together, these perspectives motivate physics-informed machine learning methods for solving option pricing problems governed by PDEs.

2.1 *From Maximum Likelihood to Mean Squared Error*

$$P(y \mid x; w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\hat{y})^2}{2\sigma^2}\right) \tag{1}$$

$$L(w) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i-\hat{y}_i^2}{2\sigma^2}\right) \tag{2}$$

$$\ln L(w) = \sum_{i=1}^{n} \left[\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(y_i-\hat{y}_i)^2}{2\sigma^2}\right] \tag{3}$$

$$\text{Maximize:} \sum_{i=i}^{n} -\frac{(y_i-\hat{y}_i)^2}{2\sigma^2} \tag{4}$$

$$\text{Minimize:} \sum_{i=1}^{n} (y_i-\hat{y}_i)^2 \tag{5}$$

Assuming Gaussian noise on the targets and maximizing the log-likelihood is therefore equivalent to minimizing the mean squared error. This justifies MSE as the standard regression loss for neural networks, and it is the starting point for the composite PINN loss described below.

2.2 *Physics-Informed Neural Networks*

A Physics-Informed Neural Network (PINN) is a neural network $\hat{V}_\theta(S,t)$ trained to satisfy both observed data and the residual of a governing PDE. Rather than discretizing the domain on a mesh, the PDE is enforced at a set of collocation points sampled from the interior and boundaries of the domain. The total loss is a weighted sum of three terms:

$$\mathcal{L}(\theta) = \lambda_{\text{pde}}\,\mathcal{L}_{\text{pde}} + \lambda_{\text{bc}}\,\mathcal{L}_{\text{bc}} + \lambda_{\text{ic}}\,\mathcal{L}_{\text{ic}}, \tag{6}$$

where

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathcal{F}[\hat{V}_\theta](S_i,t_i)\right|^2, \tag{7}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_b}\sum_{i=1}^{N_b}\left|\hat{V}_\theta(S_i^b,t_i^b) - V_b\right|^2, \tag{8}$$

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_0}\sum_{i=1}^{N_0}\left|\hat{V}_\theta(S_i,T) - \max(S_i-K,0)\right|^2. \tag{9}$$

Here $\mathcal{F}[\hat{V}_\theta]$ denotes the PDE operator applied to the network output (the residual), V_b encodes the boundary conditions (e.g., $V(0,t) = 0$ and $V(S,t) \to S$ as $S \to \infty$ for a call), and the initial condition enforces the terminal payoff at $t = T$. The partial derivatives required to evaluate $\mathcal{F}$ are computed via automatic differentiation (Goodfellow et al. 2016).

For the Black–Scholes PDE the residual is

$$\mathcal{F}[\hat{V}] = \frac{\partial \hat{V}}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \hat{V}}{\partial S^2} + rS\frac{\partial \hat{V}}{\partial S} - r\hat{V}.$$

For the Heston PDE the residual extends to two spatial dimensions (S, v), introducing an additional variance dimension and a mixed derivative term.

2.3 Activation Functions

The choice of activation function affects both the expressivity of the network and the quality of the automatic-differentiation graph used to compute PDE residuals. Standard choices such as ReLU are not suitable for PINNs because they yield zero second derivatives almost everywhere. The hyperbolic tangent $\tanh(z)$ is a natural choice: it is smooth, bounded, and its derivatives are analytically well-behaved, which is important when the loss includes second-order terms such as $\Gamma = \partial^2 V/\partial S^2$. Other smooth activations — Swish, GELU, Softplus, Sine (SIREN) — will also be explored and compared.

3 Results

These are the results

4 Conclusion

References

Abdelmessih, Kris (Feb. 2026). *A Visual Primer for Understanding Options*. Post on X (Twitter). Guest post by KeyPaganRush. Accessed: 2026-02-14. URL: https://x.com/KrisAbdelmessih/status/2022341964180718048.

Fouque, Jean-Pierre, George Papanicolaou, K. Ronnie Sircar, and Knut Solna (2011). *Multiscale Stochastic Volatility for Equity, Interest Rate, and Credit Derivatives*. Cambridge University Press.

Furnes, Egil (2026a). *FYS5429: PINNs for Option Pricing — Code Repository*. https://github.com/egil10/fys5429. Accessed: 2026.

– (2026b). *FYS5429: PINNs for Option Pricing — LaTeX Source*. https://github.com/egil10/fys5429-overleaf. Accessed: 2026.

Goodfellow, Ian, Yoshua Bengio, and Aaron Courville (2016). *Deep Learning*. MIT Press.

Gulisashvili, Archil (2012). *Analytically Tractable Stochastic Stock Price Models*. Springer.

Prefetta, — (n.d.). *Option Prices as Probabilities*. Course material / lecture notes.

Raschka, Sebastian, Yuxi Liu, and Vahid Mirjalili (2022). *Machine Learning with PyTorch and Scikit-Learn*. Packt Publishing.

5 Appendix